

# Week 5 & 6 Technical Report

## Implementation and Testing of User Authentication and Authorization System

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**Teach2Give QA & QE Training Program**

### Overview

These two weeks focused on understanding how project management, authorization, and testing fit into software development. We learned about Agile methodology and the Scrum framework, explored Jira Software to manage tasks, worked on email integration using NodeMailer, and covered the foundations of Quality Assurance and Quality Engineering. In Week Six, the sessions introduced unit testing and integration testing, showing how to verify that individual components and connected modules work as expected.

### Training Modules Covered

The training began with Agile methodology, where we learned how the Scrum framework helps teams plan and deliver work in short sprints, with clear roles for the Product Owner, Scrum Master, and Development Team.

We then explored Jira Software to manage user stories, tasks, and bugs, learning how to create, assign, and track issues, skills we applied in our group Bug Tracking System project.

Later in the week, we covered authorization, focusing on defining and controlling user access, and used NodeMailer to send automated emails from the backend securely.

In Week 6, we transitioned to testing, starting with unit testing, where we verified that individual functions or modules behave as expected. This was followed by integration testing, which tested how different components of the application work together. We also discussed the importance of testing in ensuring software quality and reliability, tying it to the QA and QE foundations we had covered earlier.

### Practical Exercise and Project

To apply the concepts, we are continuing to develop the TodoApp, where we are progressively implementing features such as authorization, email functionality, and testing. The app serves as a hands on tool to practise backend concepts, route testing, and authentication.

In parallel, we are also working on a group project, a Bug Tracking System, where Jira is used to assign and monitor tasks. Through this project, we continue to apply weekly concepts in a collaborative environment, managing workflows, integrating new features, and testing system functionality as a team.

## Challenges Faced

I initially struggled with the new topics because I hadn't fully grasped some of the concepts of the previous week, especially setting up an Express API, connecting it to SQL Server, and handling authentication using JWT. This made it difficult to follow some lesson from week five. To overcome this, I went back to redo the Week Four concepts multiple times, which helped me build a stronger understanding and confidence in starting backend projects from scratch using the modular layered architecture. Once I got comfortable with that, I was able to follow the later sessions with more ease, making Week Six much smoother.

## Recommendations

I've learned the importance of revisiting topics that I find challenging before moving forward. I recommend allocating extra practice time when dealing with complex backend setups. Using tools like Jira regularly also helps in tracking personal and group progress. Lastly, sharing challenges early with trainers or teammates can save time and keep the entire group aligned.

## Conclusion

These two weeks provided a better understanding of how backend systems are structured, tested, and managed within a team setting. By connecting earlier concepts with new ones, I improved my confidence in both development and testing processes. Relearning past topics helped me become more consistent and prepared for future projects, both individually and in collaboration with my peers.

### Signature:

David Karanja